

Sean Luka Girgis

Senior Data Engineer | Python, PySpark, AWS, ETL/ELT & Data Quality

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, PySpark/Spark, AWS data platforms, ETL/ELT pipelines, data architecture, data quality, validation, data warehousing, analytics datasets, forecasting, observability, and production troubleshooting. Strong background building scalable data workflows, transforming structured and semi-structured data, validating pipeline outputs, optimizing SQL/reporting layers, documenting operational processes, and supporting ML/AI-ready datasets in regulated enterprise environments. Best aligned to senior data engineering roles requiring Python, Spark/PySpark, AWS, ETL/ELT, data quality frameworks, data modeling, governance, stakeholder collaboration, mentoring, and analytics enablement; Snowflake, Databricks, Kafka, Airflow, MCP, Kubernetes, DataOps, and MLOps are positioned carefully as adjacent or active-learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable extraction, transformation, querying, analytics, and downstream consumption.
- Developed resilient ETL/ELT workflows for ingestion, cleansing, enrichment, standardization, validation, and delivery to analytics and business-facing reporting consumers.
- Implemented data quality and reconciliation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and trust in pipeline outputs.
- Prepared ML-ready datasets for Prophet and scikit-learn forecasting workflows, supporting capacity planning, bottleneck prediction, trend analysis, and stakeholder decision support.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Documented data flows, metric definitions, transformation rules, validation checkpoints, lineage-style notes, operational runbooks, and support procedures to improve governance and maintainability.
- Partnered with infrastructure, analytics, application, engineering, and business stakeholders to translate requirements into reliable data products and mentor teammates on reporting, validation, and operational analysis patterns.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, supporting application performance visibility and operational readiness.

- Led the FAST project, mining real-user and synthetic monitoring metrics to surface optimization opportunities adopted by engineering teams.
- Built dashboards and reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual performance-reporting workflows.
- Designed custom Management Modules, dashboards, alerts, SLA thresholds, and documentation standards used by application, middleware, infrastructure, and operations teams.
- Provided technical training, onboarding support, troubleshooting guidance, and documentation to improve adoption of monitoring standards and operational playbooks.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, regression thresholds, test findings, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java, Scala concepts

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — Capacity Forecasting Engine

Built Python/PySpark forecasting workflows to process infrastructure telemetry, prepare forecasting datasets, validate pipeline outputs, and deliver capacity and bottleneck insights through dashboards.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, BMC TrueSight, Telemetry

ServiceCall AI RAG Demo

Built staged RAG proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, retrieval decisions, answer contract schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, TF-IDF, Hybrid Retrieval, Pydantic, RAG

AI-Powered Job Search Pipeline

Built a Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking.

Technologies: Python, JSON, LLM API, FAISS, Sentence Transformers, Quality Gates